Circuit Debugging Assistant
Circuit Diagnostic Report

Report ID: ANALYSIS-6EC44782
Date: 2026-10-13 23:45
Circuit: Voltage Divider

STATUS:
POTENTIAL ISSUES IDENTIFIED

1. Uploaded versus Expected Circuit

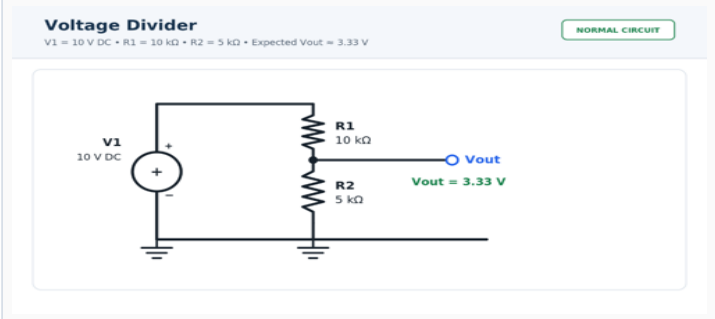


Figure 1A: Uploaded Circuit Schematic

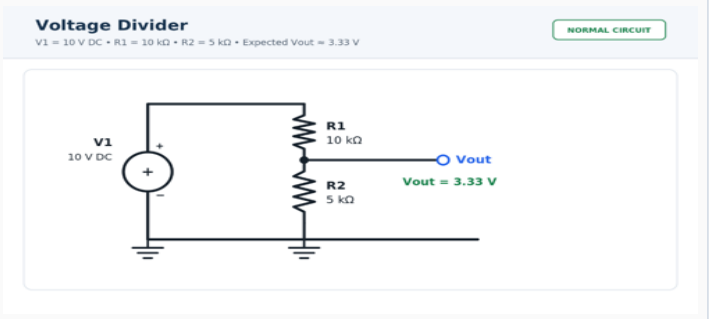


Figure 1B: Expected / Reference Circuit Model

2. Diagnosis Summary

The device under test was analyzed as a **Voltage Divider**. Numerical multi-evidence evaluation identified a probable parameter shift on **R2 +20%** (normalized residual score: **0.5622**). A total of 0 structural or parameter observations were cataloged.

3. Detected Flaws & Consistency Register

Severity	Component / Net	Finding & Evidence	Recommended Action
Major	**R2**	Normalized residual S(f)=0.562 matches R2 +20% signature.	Verify R2 value with bench meter or replace with nominal part.

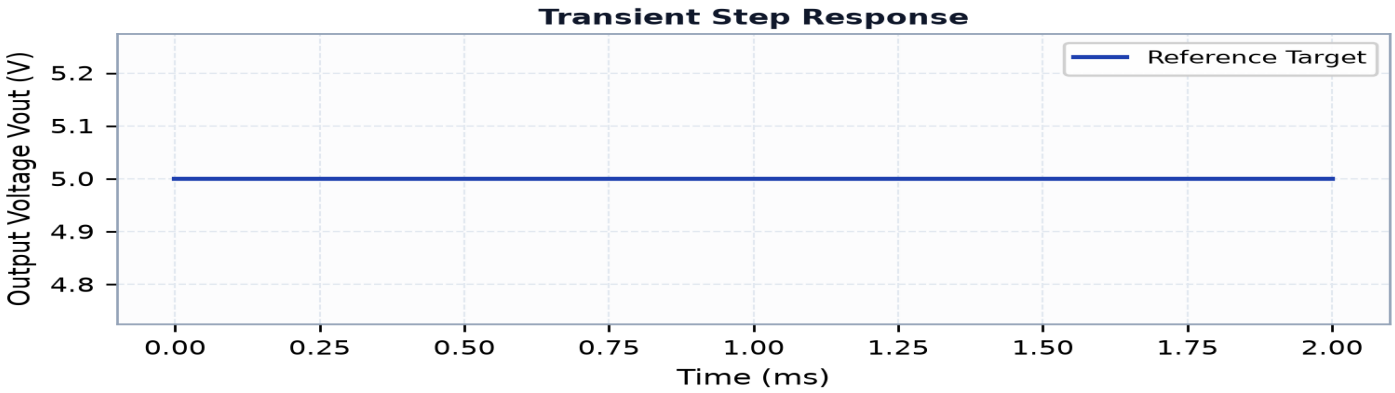
4. Confirmed Component Values

Reference	Component Type	Extracted / Nominal Value	Tolerance	Verification Status
VIN	Input Voltage Source	10 V	±1.0%	Confirmed
R1	Film Resistor	10000 Ω	±5.0%	Confirmed
R2	Film Resistor	10000 Ω	±5.0%	Confirmed

5. Expected Behaviour & Governing Calculations

• **Transfer Ratio:** Vout = Vin · [R2 / (R1 + R2)] = **5.00 V** (for Vin = 10V)
• **Total Series Current:** Is = Vin / (R1 + R2) = **0.500 mA**

6. Numerical Simulation Curves



7. Recommended Engineering Actions

1. **Physical Verification:** Perform **DMM Node Voltage (Vout)** to independently separate ambiguous parameter hypotheses.
2. **Signal Integrity:** Measure test points with a 10:1 passive probe to avoid probe capacitance loading.
3. **Post-Repair Gate:** Confirm that measured response stays within ±5% of reference cutoff before releasing design.